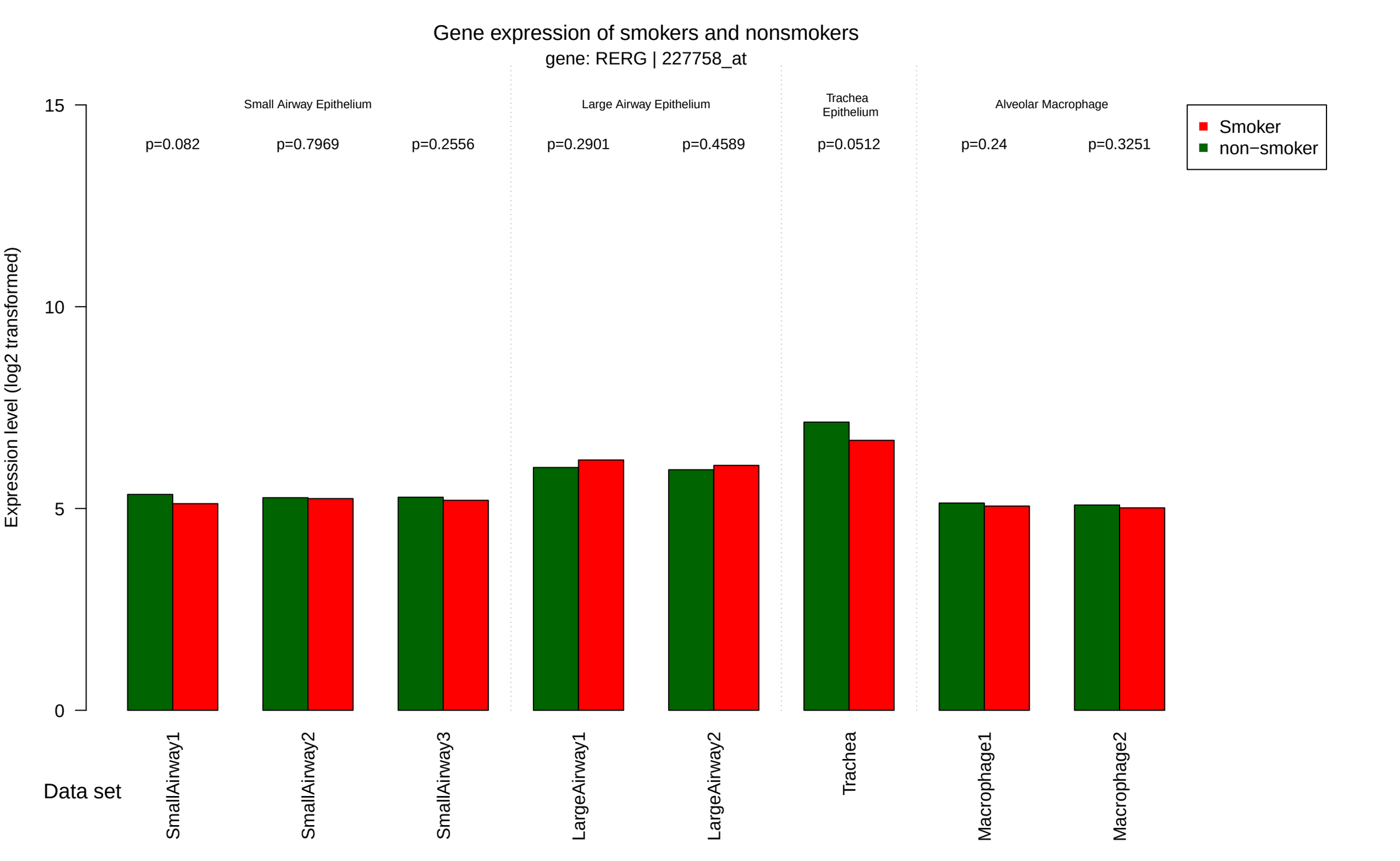
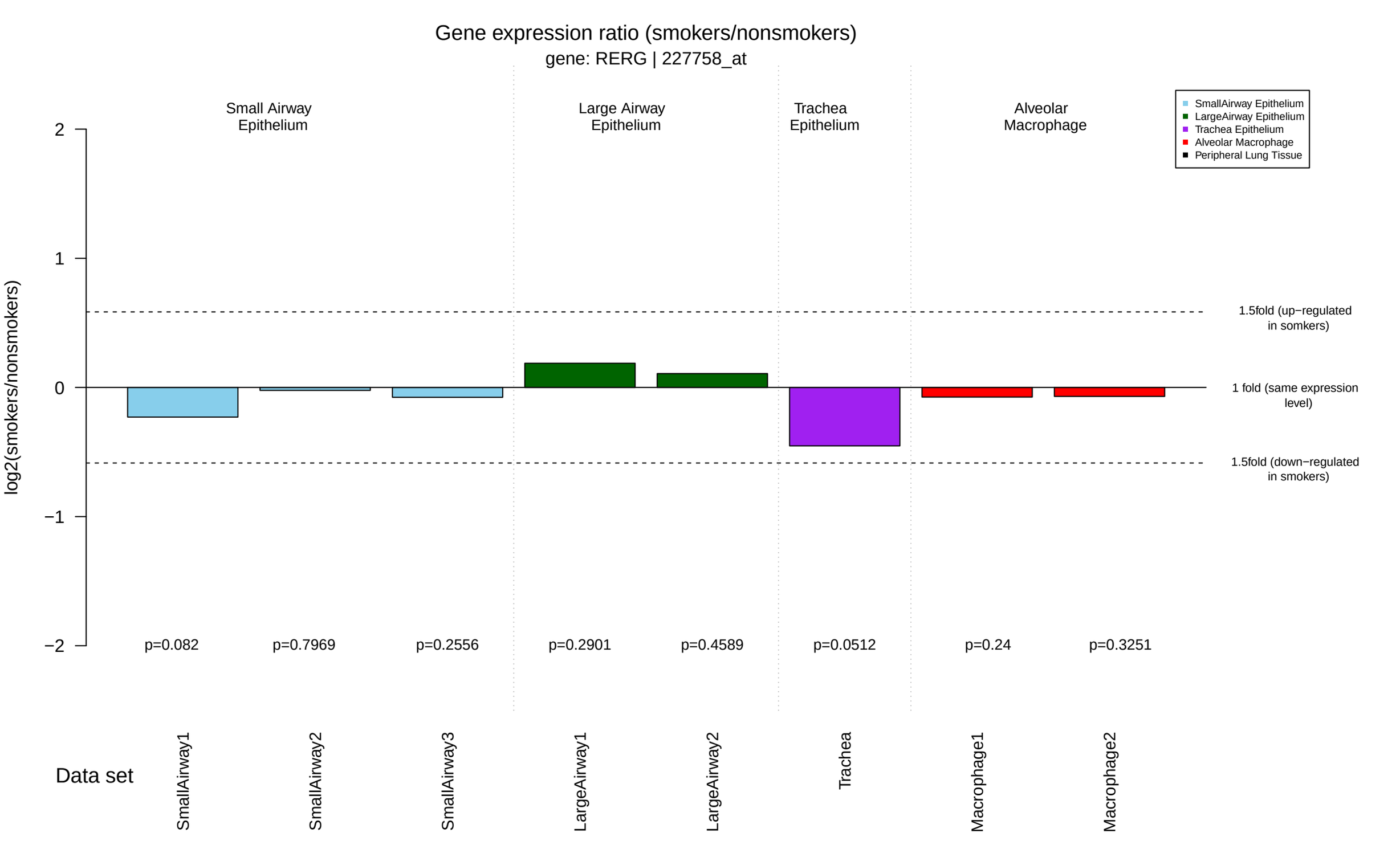
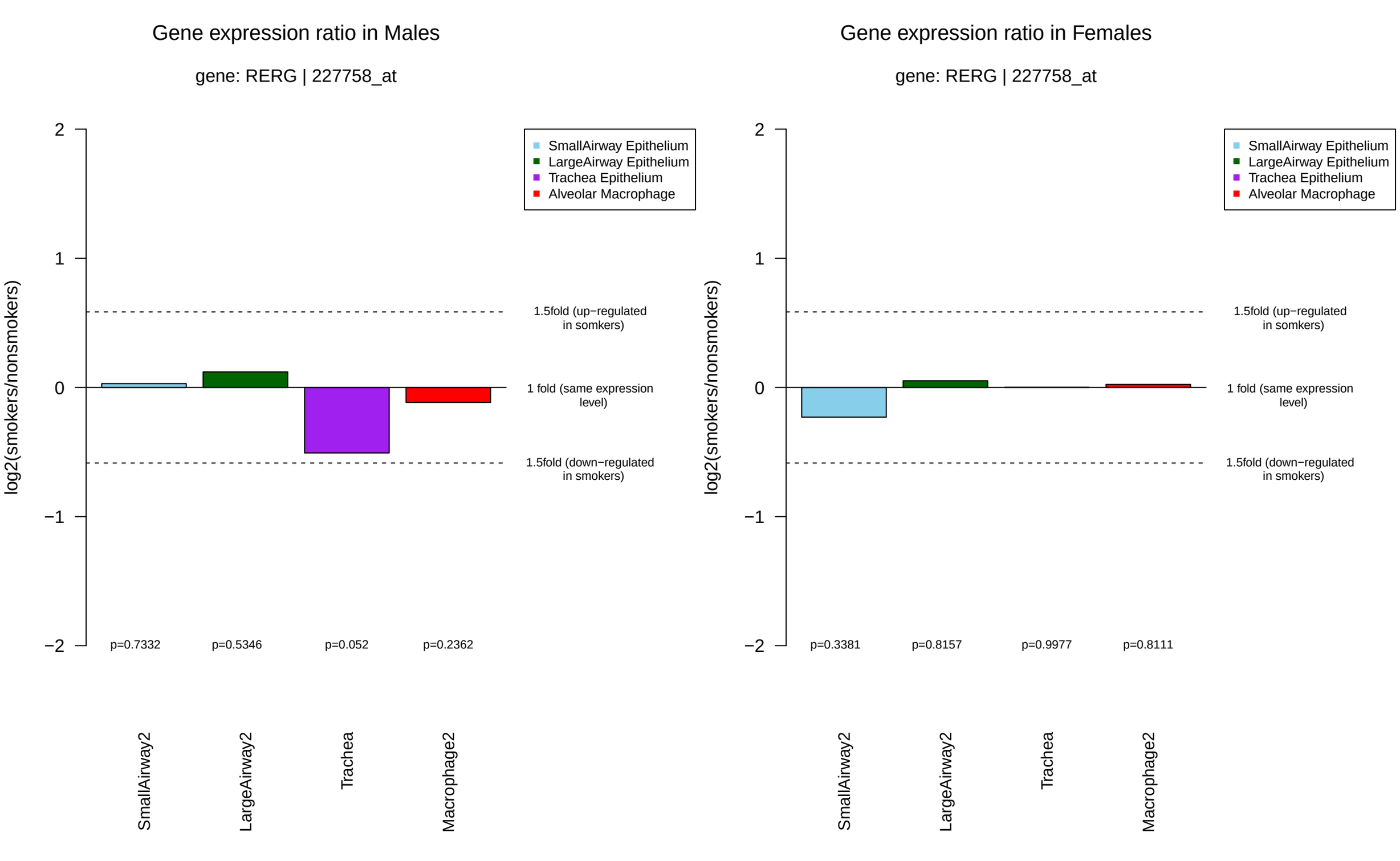
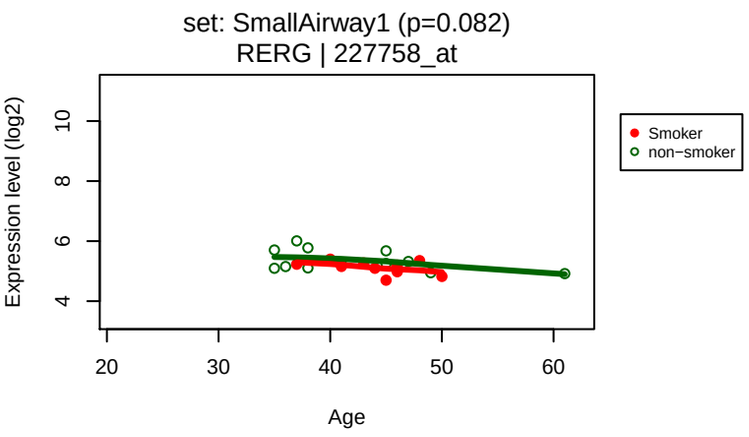
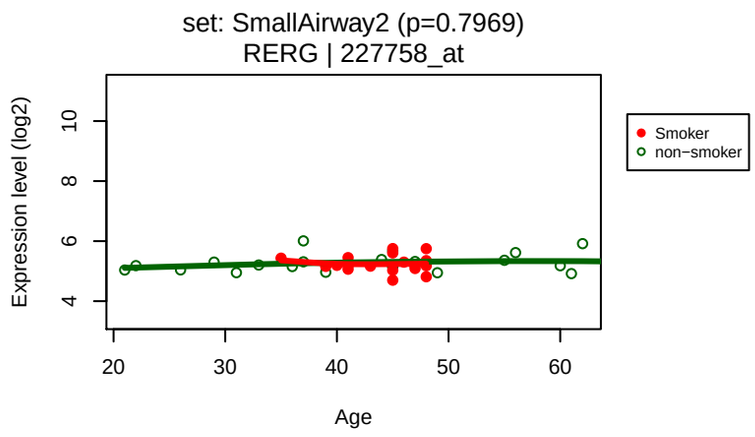
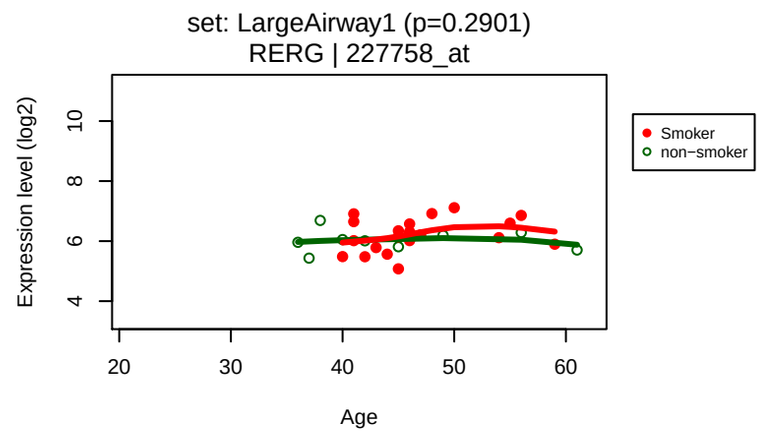

Gene expression of smokers and nonsmokers
gene: RERG | 227758_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.082
p=0.7969
p=0.2556
p=0.2901
p=0.4589
p=0.0512
p=0.24
p=0.3251
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

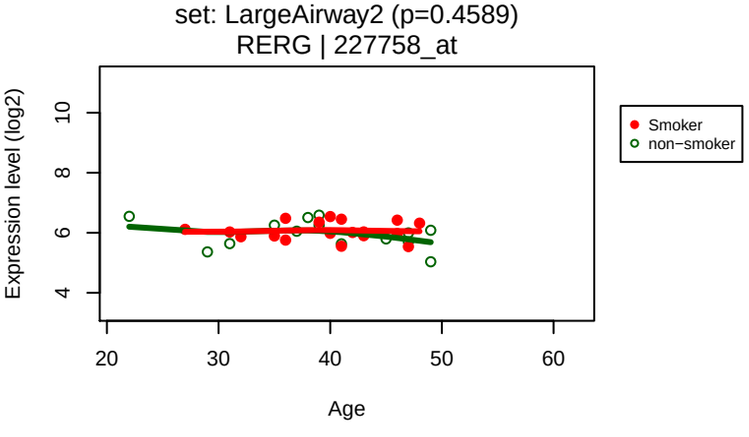

Gene expression ratio (smokers/nonsmokers)
gene: RERG | 227758_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.082
p=0.7969
p=0.2556
p=0.2901
p=0.4589
p=0.0512
p=0.24
p=0.3251
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

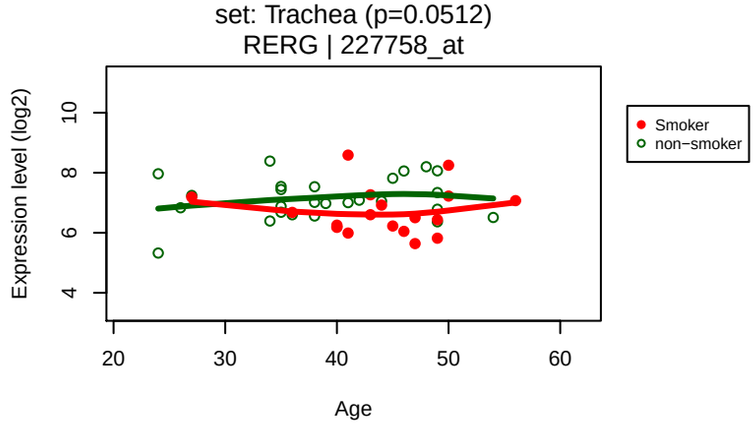

Gene expression ratio in Males
gene: RERG | 227758_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7332
p=0.5346
p=0.052
p=0.2362
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RERG | 227758_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3381
p=0.8157
p=0.9977
p=0.8111
SmallAirway2
LargeAirway2
Trachea
Macrophage2

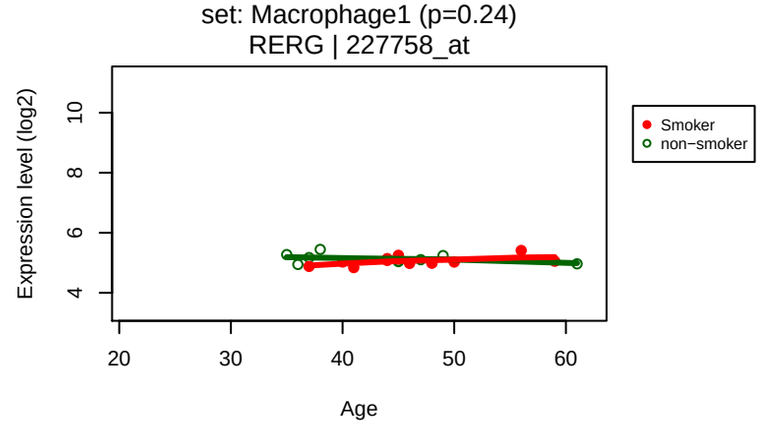

set: SmallAirway1 (p=0.082)
RERG | 227758_at
Expression level (log2)
Age
Smoker
non−smoker

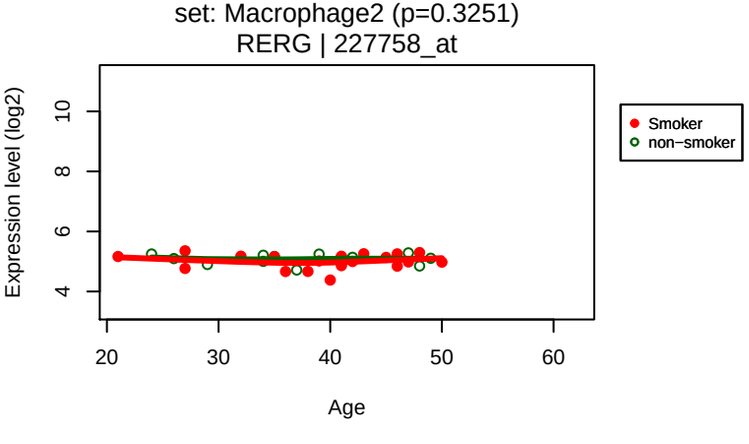

set: SmallAirway2 (p=0.7969)
RERG | 227758_at
Expression level (log2)
Age
Smoker
non−smoker

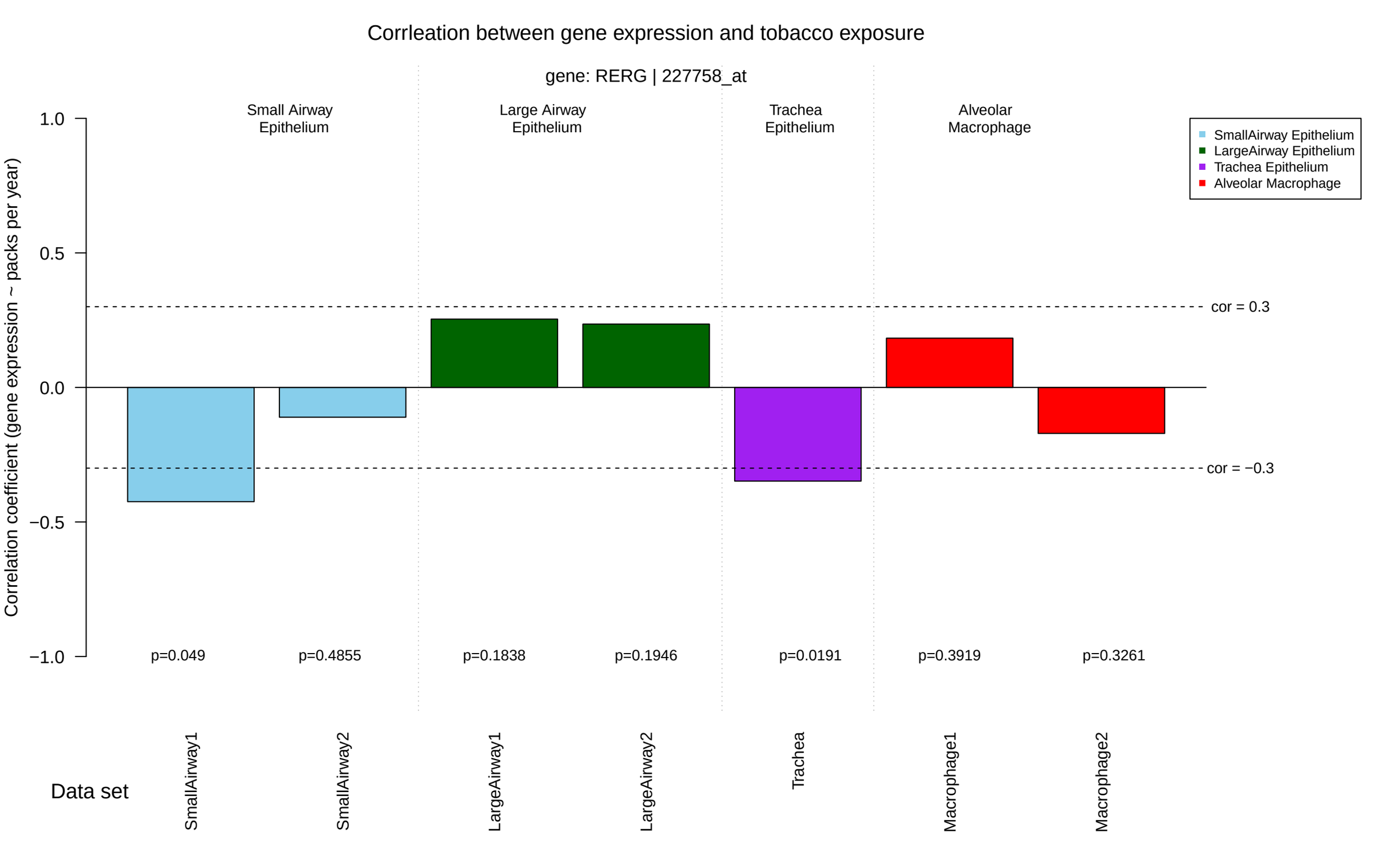

set: LargeAirway1 (p=0.2901)
RERG | 227758_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4589)
RERG | 227758_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0512)
RERG | 227758_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.24)
RERG | 227758_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3251)
RERG | 227758_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RERG | 227758_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.049
p=0.4855
p=0.1838
p=0.1946
p=0.0191
p=0.3919
p=0.3261
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2